

# Mohammad Bilal Iqbal

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## Objective

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Motivated Civil and Infrastructure Engineering student with strong analytical and design skills, proficient in AutoCAD and Civil 3D. Committed to contributing to innovative, forward-thinking structural projects by applying technical expertise and creative problem-solving. Driven to support organizational growth while continuously expanding knowledge and advancing the standards of civil engineering excellence.

## Education

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**Virginia Tech**  
*Bachelor of Science in Civil and Environmental Engineering*

**Blacksburg, VA**  
*Jan 2026 - May 2028*

**George Mason University**  
*Bachelor of Science in Civil and Infrastructural Engineering*  
3.77/4.0

**Fairfax, VA**  
*Aug 2024 - Dec 2025*

- **Awards:** Mason Distinction Award, Fort Bend ISD Academic Honors Award
- **Relevant Coursework:** Intro to Civil Engineering, Hydraulics, Civil Engineering and Architecture, Transportation Engineering, Intro to Infrastructure Systems, Geomatics, AutoCAD, Fusion360, Civil 3D, Revit, OSHA Safety Standards, OSHA Certified
- **Technical Skills:** Fusion 360 (Proficient), AutoCAD (Proficient), Civil 3D (Proficient), Hydraulics (Experienced), Civil Engineering Surveying (Experienced), Transportation Engineering Design (Knowledgeable), Environmental Engineering (Proficient), Circuit Design (Experienced), Construction Management (Experienced), Geotechnical Engineering (Experienced)
- **Programming Languages:** MATLAB, AutoCAD Revit

## Work Experience

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**STAR-TIDES / C-RASC at GMU**

**Fairfax, VA**

*Data Analyst*

*Aug 2024 - July 2025*

- Collaborated with the Institute for Sustainable Earth and STAR-TIDES under Dr. Linton Wells to support research on hurricane relief, global infrastructure resistance, sustainability, and grants and humanitarian logistics.
- Collected datasets related to sustainable technologies and federal grants and loans, emergency relief systems, and civil engineering coordination in North Carolina and Washington D.C.
- Developed data visuals, and policy briefs to propose complex resiliency plans tied to hurricane damage and civil reform.

**Shotted Specialty Coffee**

**Sterling, VA**

*Barista - Shift Lead*

*Nov 2024 - Present*

- Supervise daily cafe operations, including iced/hot drink preparations, inventory count and control, and order fulfillment
- Trained and mentored new baristas, reiterating the standards of consistency, efficiency, and safety under OSHA standards

**Coffee Lab**

**Blacksburg, VA**

*Barista*

*Jan 2026 - Present*

- Had technical experience with the espresso machine, creating drinks for customers and building a barista-customer relationship

## Academic Projects

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**Water Powered Gear-trained Hydrogen Car**

- Designed and created a functional car built with Vex Robot parts and a custom designed code and motor circuit.
- Used elements of hydrology and gear mechanisms to power the car through electrolysis.
- Collaborated with team members to problem-solve and rectify design flaws for efficient performance.

**Gear and Pulley Weight System**

- Used Vex Robot parts to create an eight-linked gear train with pulleys to lift a weight off the surface.
- Calculated tension and compression of the pulley system to mitigate gravity and external forces acting on the machine.
- Focused mathematics and physics, and practical gear rotation design, working to refine the gears and pulleys to maximum efficiency.

**Computer Powered Elevator with Raspberry Pi LED and Buzzer Control**

- Powered a motor to traverse up and down with a click of a button chosen through computer-written code and distance sonars.
- Used the Raspberry Pi LEDs to indicate the movement of the elevator, rectifying issues related to the code along the way.

**Infrastructure of a Neighborhood Design on Fusion 360**

- Created a network of roads and streets with different elevations and terrains to visualize the layout of a neighborhood
- Simulated roads, water and sewage pipes, and electricity lines to function a fully designed house externally and internally.

**Highway Physics and Calculus Analysis**

- Used technical skills in Calculus I and II as well as Physics to draw and map out a fully functional highway.
- Calculated deflecting angles on the highway to create an optimal curve based off the horizontal elevation.